LEDGER · 2026-09-04 · MATERIALS SCIENCE / CHEMICAL ENGINEERING

Supramolecular Tannic Acid-Zinc (TA-Zn) Antimicrobial Surface Shield via Ambient Aqueous Dual-Chamber Assembly

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark	Zoono Microbe Shield
Entry application	Hospital-Grade & Transit 30-Day Surface Antimicrobial Coatings
Measured advantage	< 5s surface adhesion time vs ~600s for Zoono; 4-16x MIC improvement over uncomplexed Zn baseline (not Zoono)
Time to first revenue	12 months
Gross margin at parity	98.5%
Startup CapEx	$60,000
Payback from first sale	0.0 months
Capital productivity	676.32x

Regulatory risk	Moderate — qualification costs reported ($120,000 estimated)
Market absence	Verified — the search is reported in the dossier: channels searched, companies searched, closest substitutes named

Why it failed

- Voided by: Mismatched comparison, Negative control / sub-2x
- Evidence rests on non-peer-reviewed sources

Assessment

The concept fails superiority and like-for-like comparison because antimicrobial efficacy (MIC) is benchmarked against an uncomplexed raw zinc baseline rather than the named market leader (Zoono). Furthermore, no head-to-head quantitative data is provided comparing 30-day residual microbial reduction against Zoono on surfaces.

Also flagged

- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts